LAST MINUTE REVISION ⏰

DATA ENGINEERING

SCD MASTER GUIDE FOR INTERVIEWS

Concepts + Real Examples

1. What is an SCD?
2. SCD Type 0 (Fixed)
3. SCD Type 1 (Overwrite)
4. SCD Type 2 (History)
5. SCD Type 3 (Previous)
6. Summary Matrix

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START ➡️

WHAT IS AN SCD?

Handling Data Changes

Concept.txt

```
Scenario:
An employee gets promoted.
A product price changes.

Question: How do we store this change?
A: Forget the old value?
B: Keep a history log?
C: Keep only the previous value?

-- SCDs define this strategy.
```

🌍 SCDs (Slowly Changing Dimensions) are design patterns used in Data Warehousing to manage how data changes over time.

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SCD TYPE 0

Fixed Attribute

BEFORE:

ID	Name	Hire_Date
101	John	2020-01-01

AFTER:

ID	Name	Hire_Date
101	John	2020-01-01

Ignore_Update.sql

```sql
-- Attempting to update Hire Date:
UPDATE Employees
SET Hire_Date = '2025-01-01'
WHERE ID = 101;

-- RESULT: UPDATE IS IGNORED.
```

🔒 **Immutable.** Use Type 0 for data that should NEVER change, like Original Hire Date or Date of Birth.

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SCD TYPE 1

Overwrite

BEFORE:

ID	Product	Price
500	Mouse	$50

AFTER (HISTORY LOST):

ID	Product	Price
500	Mouse	$45

Overwrite.sql

```sql
-- Simply update the value
UPDATE Products
SET Price = 45
WHERE ID = 500;

-- Warning: We lost the $50 record!
```

Overwrite. No history is kept. If you run a report for "Last Year's Revenue", it will incorrectly use the NEW price ($45).

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SCD TYPE 2

New Row (History)

Scenario: Sarah moves London -> Paris

AFTER (New Row Added):

ID	Name	City	Start	End	Active
1	Sarah	London	2020	**2024**	**N**
1	Sarah	**Paris**	**2024**	NULL	**Y**

Version_Control.sql

```sql
-- 1. Close old row
UPDATE Tbl SET Active='N', End=NOW() WHERE ID=1;

-- 2. Insert new row
INSERT INTO Tbl (ID, City, Active, Start)
VALUES (1, 'Paris', 'Y', NOW());
```

Gold Standard. Tracks full history by adding rows. "London" record is closed (Active=N), "Paris" record is opened.

SCD TYPE 3

Previous Column

BEFORE:

ID	Cur_Dept	Prev_Dept
99	Sales	NULL

AFTER:

ID	Cur_Dept	Prev_Dept
99	Mktg	Sales

Shift_Column.sql

```sql
-- Move 'Current' to 'Previous'
UPDATE Employees
SET Prev_Dept = Cur_Dept,
    Cur_Dept = 'Marketing'
WHERE ID = 99;
```

⏪ **Limited History.** Uses an extra column to store *only* the last value. You can't see where they worked 2 jobs ago.

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QUICK SUMMARY

Cheat Sheet for Interviews

Type	Action	History?
Type 0	Ignore Changes	❌ None
Type 1	Overwrite	❌ None (Current only)
Type 2	Add Row	✅ Full History
Type 3	Add Column	⚠️ Limited (1 step)

💡 **Tip:** In 90% of real-world Data Warehouses, you will use **Type 1** (for corrections) and **Type 2** (for tracking history).

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